CHAPTER 8

# Random variables and dispersion

**Chapter path**

**Objectives.** Build a distribution; interpret events; calculate expectation, variance, standard deviation; use linearity; simulate samples.
**Prerequisites.** Probabilities, weighted mean, squares, trees and lists.
**Yearly reference point.** 19-28 January: 8 hours

## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Calculate the mean of 2, 4, 6.

2. Check whether $\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)$ can be a distribution.

3. Calculate $(2-5)^2$.

4. If $P(A) = 0{,}6$, calculate $P(\overline{A})$.

5. A gross gain of 8 units and a stake of 3 give what net gain?

6. Calculate $\sqrt{0{,}25}$.

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

**Oral work and logic.** Explain why a theoretical mean can be an impossible gain. Present the difference between a probability distribution, a simulation and a statistical series.

## Investigate before formalising

Create two games with three equally likely outcomes having expectation 0 but different standard deviations. Check their distributions and explain the difference.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

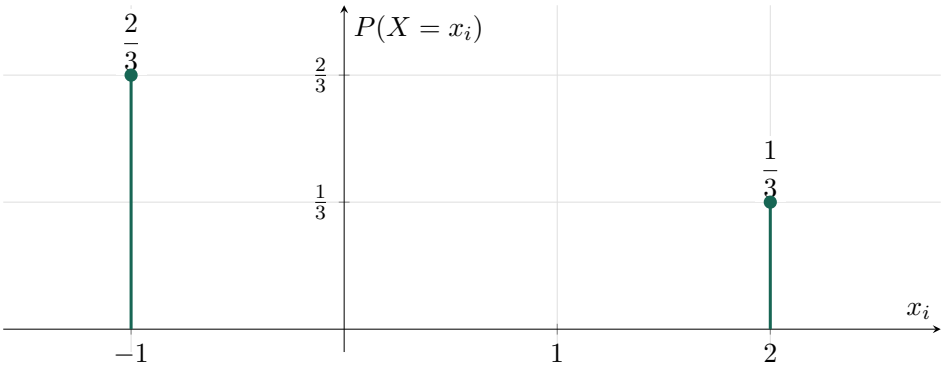
## 8.1 Variable, event and distribution

**Definition and results**

On a finite sample space Omega, a real random variable X is a function that assigns a real number to each outcome. The event $\{X = a\}$ groups the outcomes giving the value a. Its distribution assigns to each distinct value $x_i$ the probability $p_i = P(X = x_i)$, with $p_i \geq 0$ and $\sum p_i = 1$.

**Status and justification.** Definitions. The events associated with distinct values form a partition of the sample space; this justifies adding the probabilities.

**Worked example.** A fair die gives a gain X equal to 2 for 1 or 2, and to $-1$ otherwise. The distribution is $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$. $\{X \leq 0\} = \{3; 4; 5; 6\}$ : probability $2\frac{}{3}$.




The height indicates the probability: $P(X = -1) = \frac{2}{3}$ and $P(X = 2) = \frac{1}{3}$. The exact probabilities appear on both bars.

**Watch out**

The values are not always equally likely even if the outcomes are. Group the outcomes that give the same gain.

**Immediate application.** Two fair coins are tossed. X counts the tails. Write its distribution and $P(X \leq 1)$.

## 8.2 Expectation and linearity

**Definition and results**

The expectation is $E(X) = \sum_i x_i p_i$. It describes the theoretical long-run mean, not an outcome necessarily possible on a single trial. For variables X and Y on the same finite sample space and real numbers a,b: $E(aX + b) = aE(X) + b$ and $E(X + Y) = E(X) + E(Y)$, with no assumption of independence.

**Status and justification.** Summing over the outcomes $\omega$, $E(X + Y) = \sum_\omega (X(\omega) + Y(\omega))P(\omega)$ ; distributing the sum gives the result. The constant b has expectation b because the probabilities sum to 1.

**Worked example.** For the previous distribution $P(X = 2) = \frac{1}{3}$, $P(X = -1) = \frac{2}{3}$, $E(X) = 0$ : the game is fair if X is the net gain. For $Y = 3X - 2$, $E(Y) = 3 \times 0 - 2 = -2$.

**Watch out**

An expectation is not an individual guarantee; a game can lose over every time span observed despite a positive expectation.

**Immediate application.** A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair?

## 8.3 Variance, standard deviation and König·Huygens

**Definition and results**

Denoting $\mu = E(X)$, $V(X) = E((X - \mu)^2) = \sum_i (x_i - \mu)^2 p_i$. The standard deviation is $\sigma(X) = \sqrt{V(X)}$. The König-Huygens formula gives $V(X) = E(X^2) - E(X)^2$. A variance is always positive or zero; the standard deviation has the same unit as X, the variance a squared unit.

**Status and justification.** Expand $(X - \mu)^2 = X^2 - 2\mu X + \mu^2$, apply linearity and $E(X) = \mu$ : $E(X^2) - 2\mu^2 + \mu^2$. We get the formula.

**Worked example.** For X with values 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$, $E(X) = 1$, $E(X^2) = 4$, $V(X) = 3$ and $\sigma = \sqrt{3}$. Direct calculation: $(0 - 1)^2(\frac{3}{4}) + (4 - 1)^2(\frac{1}{4}) = 3$.

**Watch out**

$E(X^2)$ is not $E(X)^2$. A negative computed variance signals an error or a very small rounding error to be dealt with explicitly.

**Immediate application.** For X with values $-1$, 1 equally likely, calculate the expectation, variance and standard deviation.

## 8.4 Transforming and comparing gains

**Definition and results**

For real a, b, $V(aX+b) = a^2V(X)$ and $\sigma(aX+b) = |a|\sigma(X)$. A translation moves the mean without changing the spread; a multiplication changes the scale. Two games with the same expectation can have very different spreads.

**Status and justification.** $aX+b-E(aX+b) = a(X-E(X))$. Squaring and then taking the expectation, we get $a^2V(X)$. The square root of $a^2$ is $|a|$.

**Worked example.** Game A always pays 2. Game B pays 0 or 4 with equal probability. The two have expectation 2; variances 0 and 4. Their distributions are therefore not interchangeable despite the same mean.

**Watch out**

The variance of a sum is not, in general, the sum of the variances; no rule of this kind is used without hypotheses.

**Immediate application.** $E(X) = 3$, $V(X)$ = 4. Give the parameters of $Y = -2X + 5$.

## 8.5 Samples and the observed mean

**Definition and results**

Here, a sample of size n is a list of n independent realisations of the same variable X. Its observed mean m varies from one sample to another. We compare m with $\mu = E(X)$; repetitions make it possible to observe the proportion of samples satisfying $|m-\mu| \leq \frac{2\sigma}{\sqrt{n}}$. This threshold is an experimental task in the curriculum, not a claim of an exact coverage rate.

**Status and justification.** Experimentation: neither a convergence theorem nor the normal distribution is proved here. We check the size of the lists, the calculation of the mean and the distinction between n and the number N of samples.

**Worked example.** For X with equally likely values 0 and 2, $\mu = 1$, $\sigma$ = 1. For n=100, threshold $\frac{2}{10} = 0{,}2$: count the means in [0 ,8; 1,2]. A result of 956 out of 1000 means an observed frequency of 0.956 for this simulation only.

**Watch out**

Do not copy a single draw n times; do not state that the proportion is necessarily 95 %.

**Immediate application.** For $\mu = 5$, $\sigma$ = 3 and n=36, give the requested experimental interval.

## Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

### Exercise 8.1 · A distribution to construct

A fair die with six faces numbered from 1 to 6 is rolled. The random variable $X$ assigns to the outcome the remainder of its Euclidean division by 3. For example, a result divisible by 3 gives $X = 0$.

1. For each of the six faces, give the value taken by $X$.
2. Present the probability distribution of $X$ in a table and verify that the sum of probabilities equals 1.
3. Calculate $P(X < 2)$, stating which values of $X$ are selected.

### Exercise 8.2 · Two variance calculations

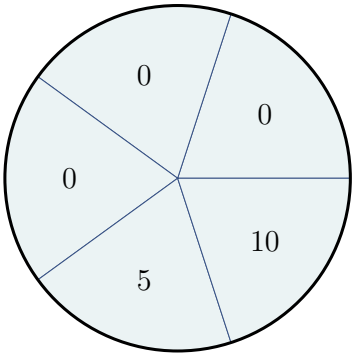
The random variable $X$ takes the values $-2$, 0 and 4 with respective probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$.

1. Calculate its expectation $\mu = E(X)$ and the moment $E(X^2)$.
2. Calculate $V(X)$ using the König-Huygens formula, then deduce the exact standard deviation.
3. Derive the variance again from the definition $V(X) = E((X-\mu)^2)$. Present the three contributions to the calculation.

### Exercise 8.3 · A fair game

A wheel has five sectors of equal angle, bearing the numbers 0, 0, 0, 5 and 10. We assume the five sectors are equally likely. The number obtained represents a gross gain $G$, in monetary units. To play, you pay a stake $m$ ; the net gain is therefore $X = G - m$.

1. Determine the distribution of $G$, then determine the stake $m$ which makes the game fair, i.e. such that $E(X) = 0$.
2. After studying variance, calculate the standard deviation of the gross gain. Deduce that of the net gain when the stake is fair.
3. Explain why changing the stake changes the expected net gain but not its standard deviation.




The five sectors have the same angle. The three sectors showing 0 are distinct and must all be counted.

### Exercise 8.4 · Same mean, different risk

Two fictional investments are offered. The first gives a certain gain $X = 1$. The second provides a gain $Y = -2$ with probability $\frac{1}{2}$ and a gain $Y = 4$ with probability $\frac{1}{2}$. The gains are expressed in the same currency unit.

1. Compare the expectations of the two gains, then their variances.

2. A person states: "These investments are equivalent since they have the same mean." Explain what this sentence leaves out and what the variance allows us to distinguish here.

### Exercise 8.5 · Linearity without independence

A fair die numbered from 1 to 6 is rolled. Let $X$ be the result of the roll and let $Y = 7 - X$. The two variables are therefore obtained from the same roll.

1. Calculate $E(X)$ and $E(Y)$, then $E(X + Y)$. Check the linearity of expectation on this example.

2. Are the variables $X$ and $Y$ independent? Justify using, for example, the events $\{X = 1\}$ and $\{Y = 6\}$.

3. Does the linearity of expectation require independence? Conclude based on your results.

### Exercise 8.6 · A minimum loss

A random variable $X$ takes finitely many values and has an expectation $\mu$. For a real number $a$, we measure the mean squared error associated with the constant prediction $a$ by $E((X - a)^2)$.

1. Prove that $E((X - a)^2) = V(X) + (a - \mu)^2$. You may write $X - a = (X - \mu) + (\mu - a)$.

2. Deduce the value of $a$ which minimises this error, then the minimum value. Justify the uniqueness of the optimal choice.

### Exercise 8.7 · Letter frequencies

We want to measure the frequency of the letter "a" in a text. The denominator is the number of alphabetic characters: spaces, digits and punctuation marks are excluded. Capital letters are treated as lower case; accented letters remain distinct from "a".

1. Write a Python function that returns this frequency. State the result chosen when the text contains no letters.
2. Test the function with `Aba !`, then with `123 !` and with the empty string. Explain the results expected before running the program.

### Exercise 8.8 · Delayed review

We repeat twice, independently, an experiment whose probability of success is $\frac{1}{2}$. The random variable $X$ counts the number of successes obtained.

1. Derive the distribution of $X$ using the four possible sequences of outcomes. Do not use the general binomial distribution formula.
2. Calculate $E(X)$ and $V(X)$ starting from this distribution.

**Independent work.** Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

## To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, Exercise 28 p. 306, all questions.** Calculate the expectation from the distribution table; keep the probabilities associated with each value.

**CQFD, exercise 31 p. 306, all questions; use the distribution from exercise 28.** Calculate the variance then the standard deviation, rounding to one decimal place as required by the textbook.

## MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. A probability distribution has probabilities summing to

   A. $0$
   B. $1$
   C. $n$
   D. the mean

2. X is 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$; $E(X) =$

   A. $1$
   B. $2$
   C. $3$
   D. $4$

3. $V(X)$ is

   A. $E(X^2)$
   B. $E(X)^2$
   C. $E(X^2) + E(X)^2$
   D. $E(X^2) - E(X)^2$

4. If $V(X) = 9$, $\sigma(X) =$

   A. $81$
   B. $9$
   C. $3$
   D. $\frac{1}{3}$

5. $E(2X + 1)$ is

   A. $2E(X)$
   B. $2E(X) + 1$
   C. $E(X) + 1$
   D. $E(X)^2 + 1$

## MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $V(X + 5)$ is

   A. $V(X) + 5$
   B. $V(X) + 25$
   C. $V(X)$
   D. $5V(X)$

2. $\sigma(-3X)$ is

   A. $3\sigma(X)$
   B. $-3\sigma(X)$

C. $9\sigma(X)$

D. $\sigma(X)$

3. The expectation is necessarily

A. a possible value

B. an integer

C. positive

D. none of these three statements

4. $E(X+Y)=E(X)+E(Y)$ requires

A. independence

B. no independence

C. X=Y

D. X positive

5. For $\sigma = 2$, n=100, $\dfrac{2\sigma}{\sqrt{n}} =$

A. $4$

B. $0{,}04$

C. $0{,}4$

D. $2$

## True or false? Justify.

1. A variance may be negative.

2. Two random variables with the same expectation have the same distribution.

3. Adding a constant to a variable does not change its standard deviation.

**Personal review**

I build a complete distribution; I distinguish gross and net; I check both variance formulas; I do not confuse frequency with probability.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

# Solutions and teaching notes, chapter 8

**Teacher use.** The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. 4. *Action:* Add, then divide by the number of values.

2. Yes: positive numbers with sum 1. *Action:* Put over a common denominator.

3. 9. *Action:* Keep the brackets before squaring.

4. 0,4. *Action:* Use the complement.

5. 5. *Action:* Write net gain = amount received minus amount paid.

6. 0,5. *Action:* Check the square of the answer.

## Investigation

Gains ( $-1$; 0; 1) and ($-3$; 0; 3) : expectation 0 in both cases, variances $\frac{2}{3}$ and 6. The second is three times more spread out in terms of standard deviation.

## Direct applications

1. Values 0,1,2 ; probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. $P(X \leq 1) = \frac{3}{4}$.

2. $E(G) = \frac{10}{5}$ = 2. For a stake m, $E(G - m) = 2 - m$ ; fair stake 2.

3. $E(X) = 0$, $E(X^2)$ = 1, variance 1 and standard deviation 1.

4. $E(Y) = -1$, $V(Y) = 16$, $\sigma(Y) = 4$.

5. $\frac{2\sigma}{\sqrt{n}}$ = 1; interval [4; 6].

## Exercises: methods and answers

**Exercise 8.1 · A distribution to construct.** Values 0,1,2; each obtained from two faces, probabilities $\frac{1}{3}$. $P(X < 2) = \frac{2}{3}$. Note the remainder 0 for 3 and 6.

**Exercise 8.2 · Two variance calculations.** $E(X) = -\frac{1}{2} + 1 = \frac{1}{2}$. $E(X^2) = 1 + 4 = 5$ ; $V = 5 - \frac{1}{4} = \frac{19}{4}$, $\sigma = \frac{\sqrt{19}}{2}$. Direct: $(25\frac{}{4})(\frac{1}{4}) + (\frac{1}{4})(\frac{1}{2}) + (\frac{49}{4})(\frac{1}{4}) = \frac{76}{16} = \frac{19}{4}$.

**Exercise 8.3 · A fair game.** $E(G) = \frac{15}{5}$ = 3, so m=3. $E(G^2) = \frac{125}{5} = 25$ ; $V(G) = 25 - 9$ = 16, standard deviation 4, unchanged by the stake.

**Exercise 8.4 · Same mean, different risk.** $E(X) = E(Y) = 1$ ; $V(X) = 0$ ; $E(Y^2) = 10$, $V(Y) = 9$. X is certain, Y is spread out. A player's preferences cannot be ranked solely from the expectation.

**Exercise 8.5 · Linearity without independence.** $E(X) = \dfrac{1+2+3+4+5+6}{6} = \dfrac{7}{2}$ ; $E(Y) = 7 - E(X) = \dfrac{7}{2}$. Since $X + Y = 7$, its expectation is 7, the sum of the two expectations. The events $\{X = 1\}$ and $\{Y = 6\}$ are identical: their intersection has probability $\dfrac{1}{6}$, while the product of their probabilities is $\dfrac{1}{36}$. The variables are not independent. Linearity still holds without this hypothesis.

**Exercise 8.6 · A minimum loss.** Write $X - a = (X - \mu) + (\mu - a)$, expand and then use $E(X - \mu) = 0$. The square $(a - \mu)^2$ is minimal, zero only for $a = \mu$. The mean minimises the mean squared deviation.

**Exercise 8.7 · Letter frequencies.** Convert the text to lower case, keep the characters for which `isalpha()` is true, then count the "a"s. Divide this number by the number of letters if this is non-zero. For a text with no letters, return `None`, since the frequency is undefined. For `Aba !`, two "a" out of three letters give $\dfrac{2}{3}$ ; the other two tests give `None`. The supplied program `frequence_a` follows this convention.

**Exercise 8.8 · Delayed review.** Distrib.: 0 , 1 , 2 with probabilities $\dfrac{1}{4}, \dfrac{1}{2}, \dfrac{1}{4}$. Expectation 1 ; moment of order 2: $\dfrac{1}{2} + 1 = \dfrac{3}{2}$, variance $1\frac{1}{2}$. The four paths are counted explicitly.

**CQFD, Exercise 28 p. 306, all questions.** $E(X) = (-10) \times 0{,}35 + (-5) \times 0{,}15 + 0 \times 0{,}4 + 20 \times 0{,}1 = -2{,}25 = -\dfrac{9}{4}$. A negative expectation is compatible with a possible positive value.

**CQFD, exercise 31 p. 306, all questions; use the distribution from exercise 28.** $E(X^2) = 100 \times 0{,}35 + 25 \times 0{,}15 + 400 \times 0{,}1 = 78{,}75$. Thus $V(X) = 78{,}75 - (-2{,}25)^2 = 73{,}6875 = \dfrac{1179}{16}$, then $\sigma(X) = \dfrac{\sqrt{1179}}{4} \approx 8{,}6$. Do not round the expectation before calculating the variance.

## MCQ: answers and reasons

**MCQ 1.**

1. B. Partition of the sample space.

2. A. $0 + \dfrac{4}{4}$.

3. D. König-Huygens.

4. C. Positive square root.

5. B. Linearity.

**MCQ 2.**

1. C. A translation adds no spread.

2. A. Absolute value of the factor.

3. D. A variable may be negative or have a mean that it never takes.

4. B. A common finite sample space is enough.

5. C. $\dfrac{4}{10} = 0{,}4$.

## True or false

1. False: it is a mean of squares with positive weights.

2. False: the constant 1 and an equally likely variable $-2,\ 4$ have the same mean.

3. True: the deviations from the new mean stay the same.

**Differentiated teaching.** If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.